

# Oxidation Numbers and Redox Reactions

Answer key with worked solutions

## A. Understanding

### 1. Definitions.

**Oxidation** is the loss of electrons; the oxidation number **increases**. **Reduction** is the gain of electrons; the oxidation number **decreases**.

## B. Oxidation numbers

### 2. Oxidation numbers.

(a) **-3** (b) **+5** (c) **+6** (d) **+6** (e) **+3** (f) **-1** (peroxide) (g) **+4** (h) **+5**. For example, (h):  $x + 3(-2) = -1$ , so  $x = +5$ .

## C. Agents and half-equations

### 3. Chlorine and bromide.

**Br<sup>-</sup> is oxidized** ( $-1 \rightarrow 0$ ) and **Cl<sub>2</sub> is reduced** ( $0 \rightarrow -1$ ). **Oxidizing agent: Cl<sub>2</sub>; reducing agent: Br<sup>-</sup>.**

### 4. Half-equations.

Oxidation:  $\text{Mg} \rightarrow \text{Mg}^{2+} + 2\text{e}^-$ . Reduction:  $2\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2$ .

### 5. Redox or not?

(a) **Not redox**: no oxidation number changes. (b) **Redox** (disproportionation): O goes from  $-1$  to  $-2$  in H<sub>2</sub>O and to  $0$  in O<sub>2</sub>. (c) **Not redox**: a precipitation in which all oxidation numbers stay the same.

## D. Balancing

### 6. Copper and silver.

Cu loses  $2 e^-$ ; each  $Ag^+$  gains  $1 e^-$ , so 2  $Ag^+$  are needed:  $Cu + 2 Ag^+ \longrightarrow Cu^{2+} + 2 Ag$ .  
Charge:  $+2 = +2$ . ✓

### 7. Dichromate and iron(II).

Reduction:  $Cr_2O_7^{2-} + 14 H^+ + 6 e^- \longrightarrow 2 Cr^{3+} + 7 H_2O$

Oxidation ( $\times 6$ ):  $6 Fe^{2+} \longrightarrow 6 Fe^{3+} + 6 e^-$

Overall:  $Cr_2O_7^{2-} + 14 H^+ + 6 Fe^{2+} \longrightarrow 2 Cr^{3+} + 7 H_2O + 6 Fe^{3+}$ . Charge:  $-2 + 14 + 12 = +24$  on the left;  $+6 + 18 = +24$  on the right. ✓

### 8. Copper and nitrate.

Oxidation ( $\times 3$ ):  $3 Cu \longrightarrow 3 Cu^{2+} + 6 e^-$

Reduction ( $\times 2$ ):  $2 NO_3^- + 8 H^+ + 6 e^- \longrightarrow 2 NO + 4 H_2O$

Overall:  $3 Cu + 2 NO_3^- + 8 H^+ \longrightarrow 3 Cu^{2+} + 2 NO + 4 H_2O$ . Charge:  $-2 + 8 = +6 = +6$ . ✓

## E. Challenge

### 9. Permanganate.

(a) Reduction ( $\times 2$ ):  $2 MnO_4^- + 16 H^+ + 10 e^- \longrightarrow 2 Mn^{2+} + 8 H_2O$ ; oxidation ( $\times 5$ ):  
 $5 C_2O_4^{2-} \longrightarrow 10 CO_2 + 10 e^-$

Overall:  $2 MnO_4^- + 5 C_2O_4^{2-} + 16 H^+ \longrightarrow 2 Mn^{2+} + 10 CO_2 + 8 H_2O$ . Charge:  $-2 - 10 + 16 = +4 = +4$ . ✓

(b)  $n(MnO_4^-) = 0.0200 \text{ mol/L} \times 0.02000 \text{ L} = 4.00 \times 10^{-4} \text{ mol}$

$n(Fe^{2+}) = 4.00 \times 10^{-4} \text{ mol } MnO_4^- \times \frac{5 \text{ mol } Fe^{2+}}{1 \text{ mol } MnO_4^-} = 2.00 \times 10^{-3} \text{ mol}$

$[Fe^{2+}] = \frac{2.00 \times 10^{-3} \text{ mol}}{0.02500 \text{ L}} = 0.0800 \text{ M}$